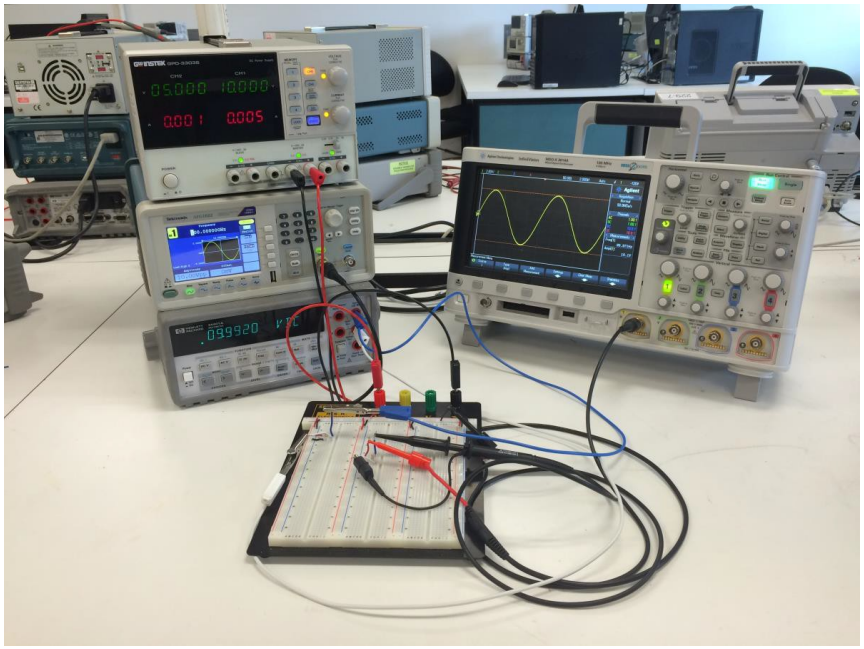
ECET 250 E Lab Equipments:

A STARTER GUIDE



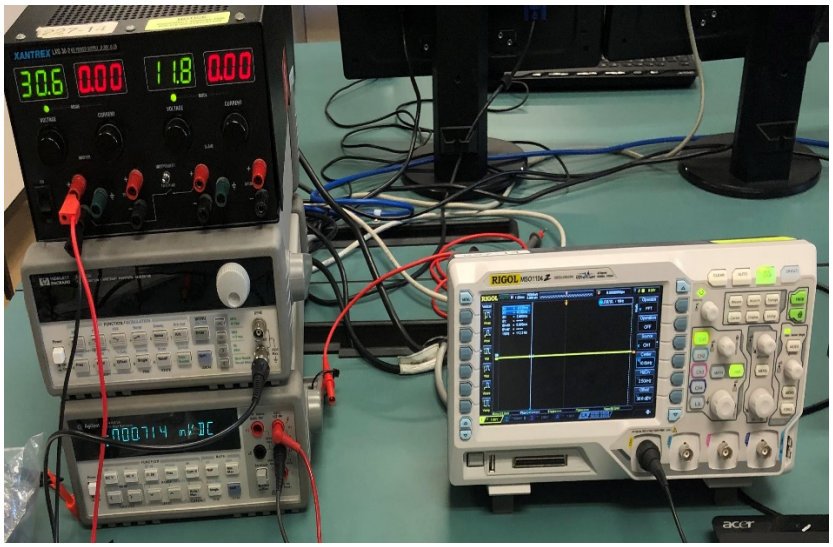


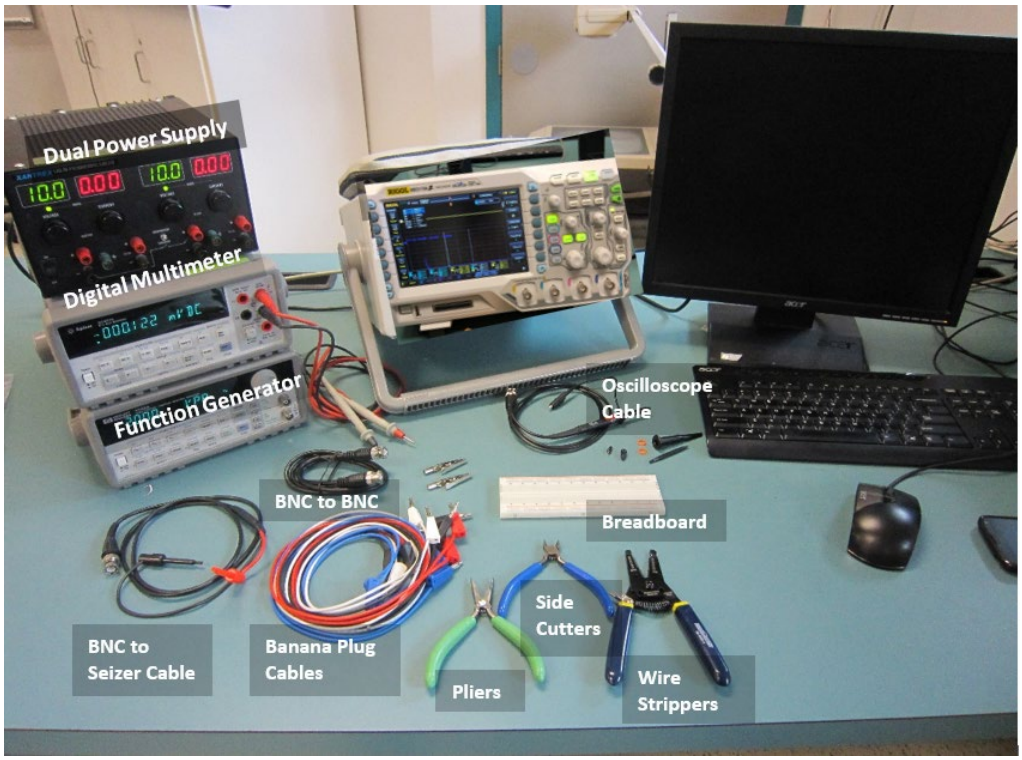
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3. Function Generator
4. Oscilloscope

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Introduction

Welcome to the Electrical and Computer Engineering program at Camosun! Many students have not had experience with the instruments used for circuit analysis before starting the program.
Lab periods are a chance to reinforce the theory learned in lectures. Developing these concepts into a practical understanding is an important part of becoming a technologist. However, having been through the first labs we know that it can be quite intimidating. With this in mind and we have put together *ECET Labs 250E: A Starter Guide*. Our intention is to cover the basics for using the lab equipment and the tools most commonly used throughout the technologist program.
Your first lab(s) are designed for you to gain familiarity with these instruments. We aim for this document to act as a reference and helping hand.



A Typical Electronics Lab Station with some of the student tool kit laid out on the bench. The student tool kit consists of (in part) a breadboard, a side cutter, a wire stripper, a needle nose pliers, a BNC to BNC cable for the signal generator, a BNC to seizer clips(+ red/ – black) cable for the signal generator, several banana plug cables in red, white, blue, and black, two alligator clips for use with the banana plugs, and two oscilloscope probes with extra attachments (only one probe shown). Note that the Digital Multimeter (DMM) is the only equipment for which probes are supplied.

1. Toolkit

The student tool and parts kit are as essential to working in the lab as the devices covered in this manual. A few of the most used cables and tools in circuit analysis are given a quick description below.

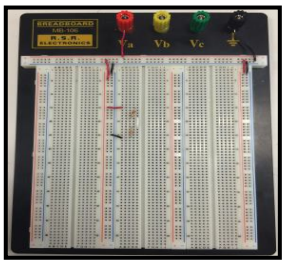


Figure1.1 Breadboard

Breadboard – The solderless breadboard is used to build, test, analyze, and experiment with circuits without the commitment of soldering anything into place. The power rails are outlined in the red (+/Vs), and blue (- / gnd). On the inside of the rails each of the five sockets are grouping called a bus. These are connected, and act as a single node. Thisboard will quickly become your third lab partner and it's definitely recommended to purchase an extra one (or three)!

Banana Plug Cables – Used mainly with the DMM and DC Power Supply in the lab, these cables will also plug into the breadboard supplied in your student kit. The cables come in a variety of colours which don't affect their use. However, we recommend keeping the conventions of red(+) and black(-) to simplify using them in labs. Also included with the cables are a pair of Alligator Clips (Fig 1.2). These removable will attached to the ends of the cables and can be used to take measurements over different parts of a circuit. We will show a few examples in the coming pages.

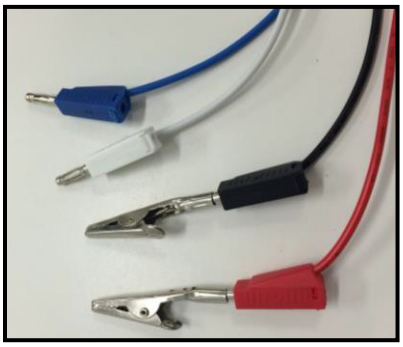


Figure 1.2 – Banana Plug Cables

BNC (Bayonet Neill–Concelman) Connector to Seizer Cables – These are primarily used to provide AC signals from the function generator to your circuit. The single-ended BNC connector plugs into the function generator with a quarter turn. The black and red seizer clips will attach to a wire inserted into the bread board. You will also have BNC to BNC cables as part of your kit. These can be used to connect directly from the function generator to the oscilloscope.



Figure 1.3 – BNC to Seizer Cables

Oscilloscope Probes – As the name suggests these probes are used with the Oscilloscope to measure and display AC signals over a circuit. The probe is generally placed at the V_{in} and the alligator clip covered in rubber at ground.Take special note of the red selector switch on the body of the probe with labels 1x, 10x and REF. It will be covered in detail with an example in the oscilloscope section.



Figure 1.4 – Oscilloscope Probes

Devices

In this section we will cover four of the main instruments at each lab station:

1. DC power supply,
2. Digital multimeter (DMM),
3. Function generator,
4. Oscilloscope

We will onlycover the basic, but essential uses for these powerful tools. It should be noted that that the equipment does vary between labs, however most devices have similar features. It is important that before using any piece of equipment its data sheet should be read to understand its limitations and design. If you are unsure about anything, ask your instructor before attempting to use a piece of equipment you are unfamiliar with. A list of the make and model of each devicecan be found in appendix A as reference for looking up data-sheets.

Note: The first step in using these devices is turning the power on and for each device we've highlighted the power button with a green square.



2.Dual Power Supply



Figure 2.1 Dual Power Supply. Note that each side displays the voltage set (Green) and current drawn (Red). Output for each is the Red (+) and Black (-) terminal below each of the displays. There are also two 5V outputs. The two adjustable power sources can be set as independent (normal use) or in tracking mode (where changes in one side are reflected in the other). Note that the left-hand supply is set at 20.0 Volts and the right-hand supply is set at 10.1 Volts. Neither supply is sourcing any current at the moment – which is good since there is nothing connected to the variable output terminals.

The lab station power supplies have dual outputs that can be varied from 0 – 30 V at 0 – 2 A either in independent or tracking mode. They also typically have either one or two additional outputs set at 5 V and 3 A. The dual supplies can be connected in series to provide up to 60 V – this should only be attempted under supervision.

There are two earth or ground connections (green terminals) that can be used as a reference for your circuit, particularly when you require positive and negative voltages (eg for an op amp circuit). These terminals connect through the building ground via the electrical outlet into which the power supply is plugged.

The desired voltage is set using the knob labelled VOLTAGE. The voltage display should not be considered accurate and you should get a more precise measurement of the voltage supplied to your circuit using the DMM. The CURRENT knob is used to provide current limiting. If you have set up a voltage and connect to a circuit that draws more current than the supply can provide then you will notice the current limit light is on (red) and the voltage will drop significantly. Whether the current is being limited or not, the amount of current draw will be indicated in the display to the right of the voltage display.

It is always a good idea to have some idea of the theoretical current draw of your circuit and use the current limiting feature of the power supply to limit the available current to this theoretical maximum. This can help prevent damage to components that can occur if your circuit has a short or an incorrect component value (eg 10 Ω resistor instead of a 10 kΩ resistor).

The power supplies currently available in our lab have overcurrent protection built in to them – some older power supplies will not have this feature. There are two ways to set up current limiting on these power supplies: the first method you can only use with power supplies that have overcurrent protection, the second method should work with any power supply that allows current limiting. For both methods lets assume that the supply voltage should be 10 Volts and the maximum current should be 10 milliamps (mA).



Figure 2.2a. *Current limiting the power supply. Method 1 for power supplies with over current protection*

Step 1: *Set up required voltage (in this case 10 volts).*



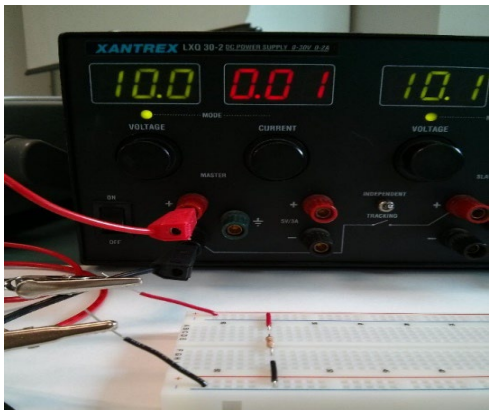
Step 2: *Turn OFF the power supply.*
Step 3: *Short the Red (+) and Black (-) terminals using a banana plug cable. Do NOT do this with an older power supply that does not have overcurrent protection – at a minimum a fuse will blow.*

Step 4: *Turn ON the power supply. Note that the voltage is 0V (not 10 V) since a non-zero voltage cannot be maintained across 0 Ω resistance (a short) and the current it limited to 0.00 Amps. The current limit lamp is lit.*



Step 5: *Adjust the current limiting knob until the desired current is displayed (in this case in red) eg 0.01 Amps = 10 mA.*

Step 6: *Turn OFF the power supply.*

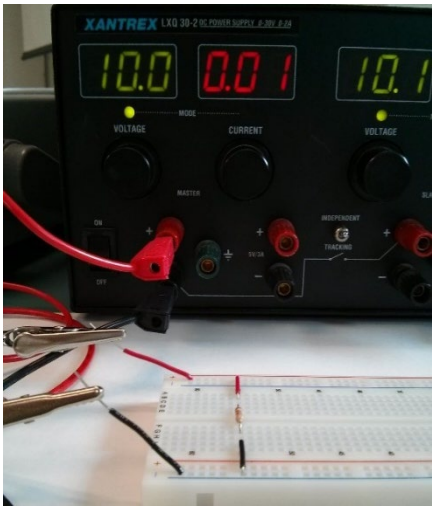


Step 7: *Connect the power supply to your circuit and turn it ON.*



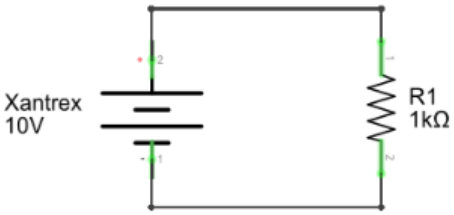
Figure 2.2*b Current limiting the power supply. Method 2 for any power supply without over current protection*

Step 1: *Set up required voltage (in this case 10 volts).*



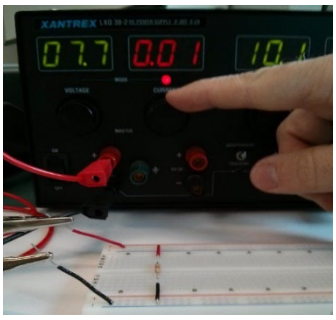
Step 2: *Set up a resistor with a value that will allow the maximum current desired. In this case, 10 V supply and a 1 kΩ resistor yield 10 mA.*

Step 3: *Turn OFF the power supply.*
Step 4: *Connect the Red (+) terminal to one end of the resistor and the Black (-) terminal to the other end of the resistor. In this case the resistor is mounted on a breadboard and is connected to the power supply with banana plug connectors and alligator clips. The effective circuit is shown as a schematic below.*



Step 5: *Turn ON the power supply. Note that it is NOT yet current limited (except by the external resistor) and is sourcing the expected (approximately) 10 milliamps.*

Step 6: *Adjust the current limit knob until the voltage displayed on the power supply just starts to drop and the red lamp lights. Now, for any circuit impedance higher than 1 kΩ current will be less than 10 mA. For a lower impedance, the current will not exceed approximately 10 mA and the supply voltage will likely drop below 10 V BUT your components will hopefully not be damaged.*



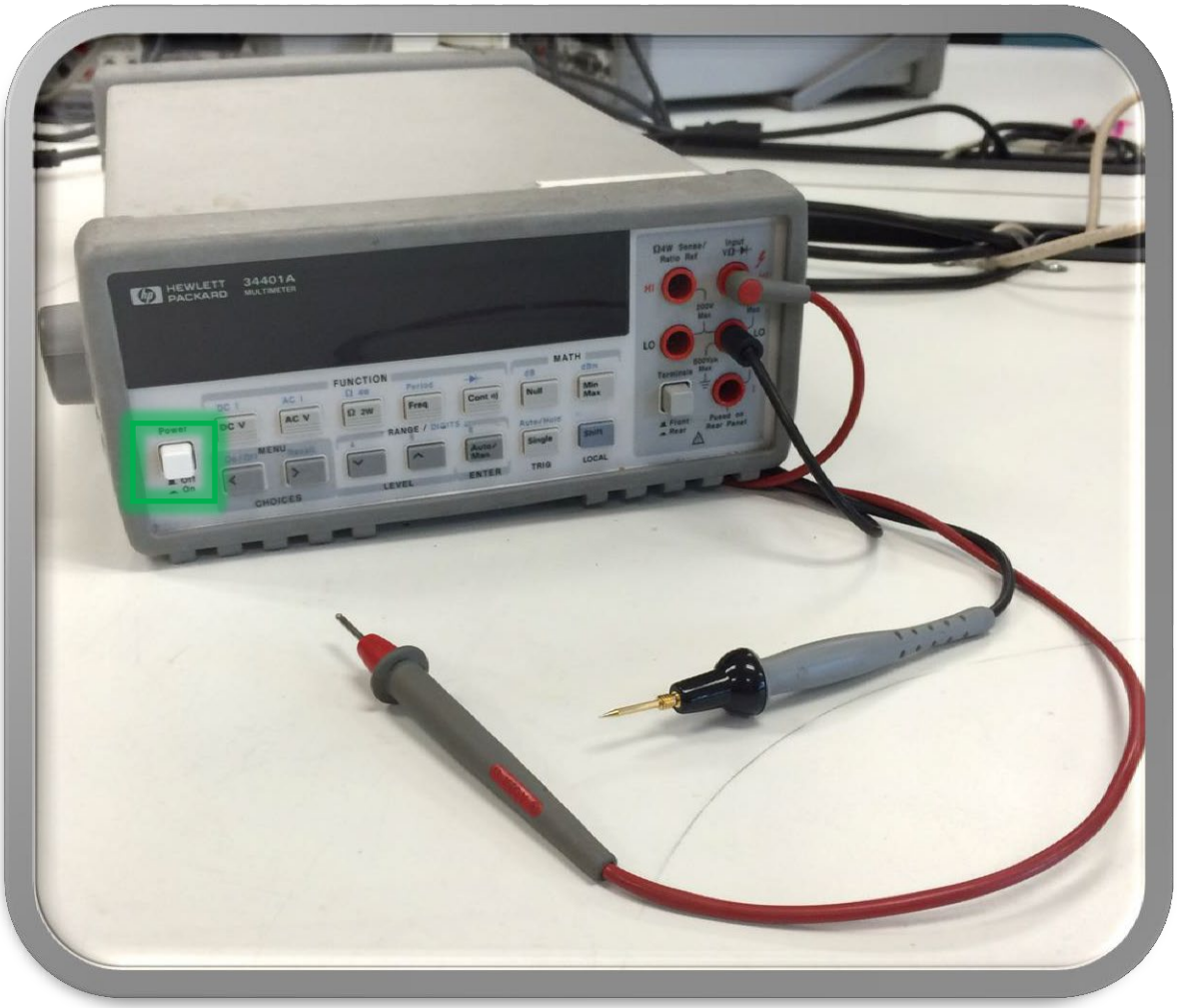
Step 7: *Turn OFF the power supply and reconnect it to your circuit, turn it back ON and watch the sparks fly.*

NOTES:

1. IMPORTANT: Do NOT stick the DMM probes into the ventilation holes on top of the Power Supply.
2. Model of Power Supply will vary from lab to lab.
3. You should look up the data sheet for the power supply at your lab station.

3. Digital Multimeter

The digital multimeter (DMM as it is usually referred to) is used to measure DC and AC voltages (V), resistance (Ω), and current (I). There are probes supplied with the DMM at each lab station for quick measurements, however it is often more convenient to use the banana plug cables mentioned in the Toolkit section.



Measuring a Resistor

This is a procedure that you will do in almost every lab. The nominal value given by the resistor's colour band is only accurate to the percentage band. To make calculations and experiments more precise you need to know the actual value of the resistor. This has the added benefit of confirming the resistor's intended value as it is a common error to confuse one resistor for another. A 1000 Ω resistor (brown, black, red), and a 10,000Ω resistor (brown black orange) are easily mistaken for one another and will likely produce very different results in your circuit.
As part of the appendices a resistor colour band chart has been provided for reference.

1. Press the **Ω 2W** button.
2. Insert the banana plug cables or probes.

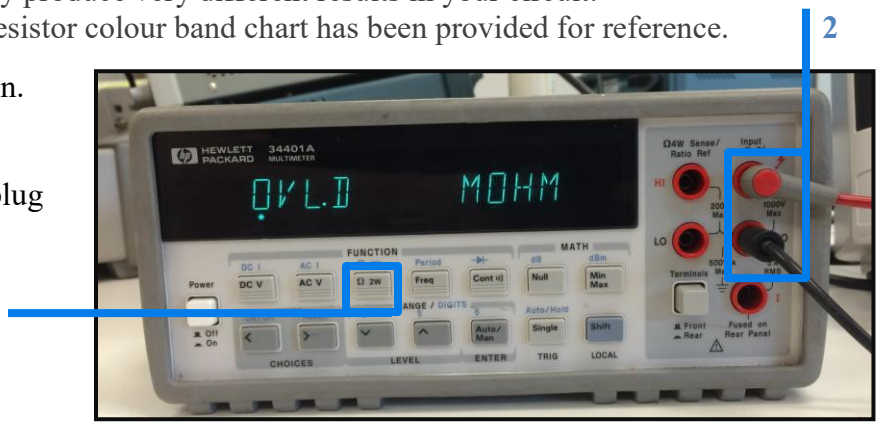


Figure 3.10 – Ohm-meter Display
The OVLD display means it is measuring an open circuit(∞ resistance).

3. Place the resistor in the breadboard(Fig. 3.11) or clamp each end with alligator clips (Fig. 3.12).

Note: If your fingers are touching more than one end of the resistor you will get an improper reading as the small current generated by the ohm meter function will partially travel through you! The two methods shown prevent this.



Figure 3.11 – Alligator Clip Method

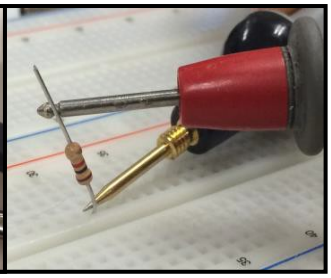


Figure 3.12 – Using theDMM Probes

The resistance will be displayed as shown (Fig. 3.13). Be careful to note the units at the end of the value as well as the decimal place.



Figure 3.13 – Two ResistanceReadings

Measuring Voltage

1. Press the **DC V** button.

2. Insert the banana plug cables.

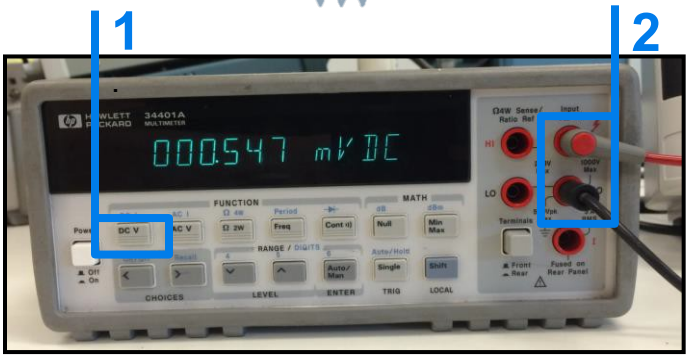


Figure 3.20 – Voltmeter Display

3. Place a wire attached to theclips into the circuit (Fig. 3.21).

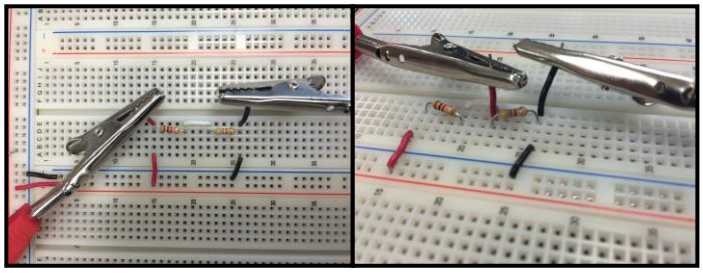


Figure 3.21 – Measuring the Voltage Over Resistors

Note: The probes attached to the DMM may also be used for quick measurements. If a measurement is to be taken over a period of time we recommend using the alligator clip/wire setup as shown.

The voltage reading will display as shown in figure 3.22. Similarly, to the resistance readings always be careful about reading decimal places and units correctly.



Figure 3.22 – Voltage Reading
The voltage will be displayed in as either positive (no sign) or negative depending on the orientation of the leads.

Measuring Current

Often in the lab current is calculated using ohms law with the measured voltage over a resistor. However, the DMM is capable of measuring 2A-3A of DC or AC current.

IMPORTANT: To measure current the DMM MUST be connected in SERIES with the circuit, not in parallel as when measuring voltage. If you are unsure about this instruction ask your instructor before proceeding with the measurement.

1. Press the **Shift** button.
2. Press the **DC I** or **AC I** button depending on the measurement needed.

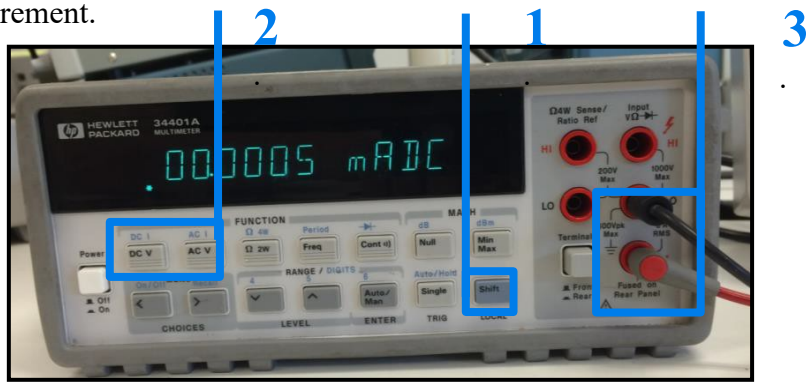


Figure 3.30 – Ammeter Display

3. Insert the banana plug cables.
4. Plug the wires attached to the alligator clips into circuit to be measured (Fig 3.31).

   Note: Without the ammeter connected this would be an open circuit. This is an example of how to use the ammeter in SERIES.

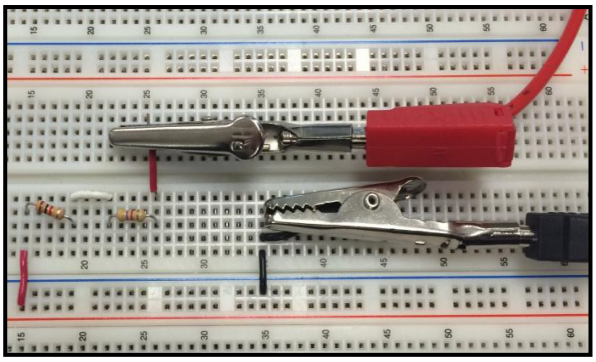


Figure 3.31 - Circuit Setup for Measuring Current

The reading will be displayed on the screen in ADC or AAC depending on the measurement type.

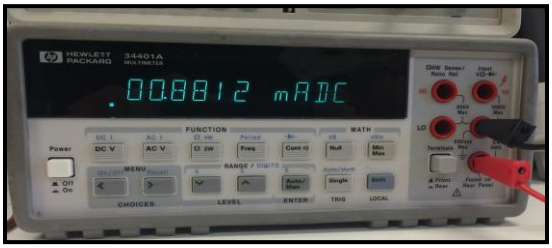


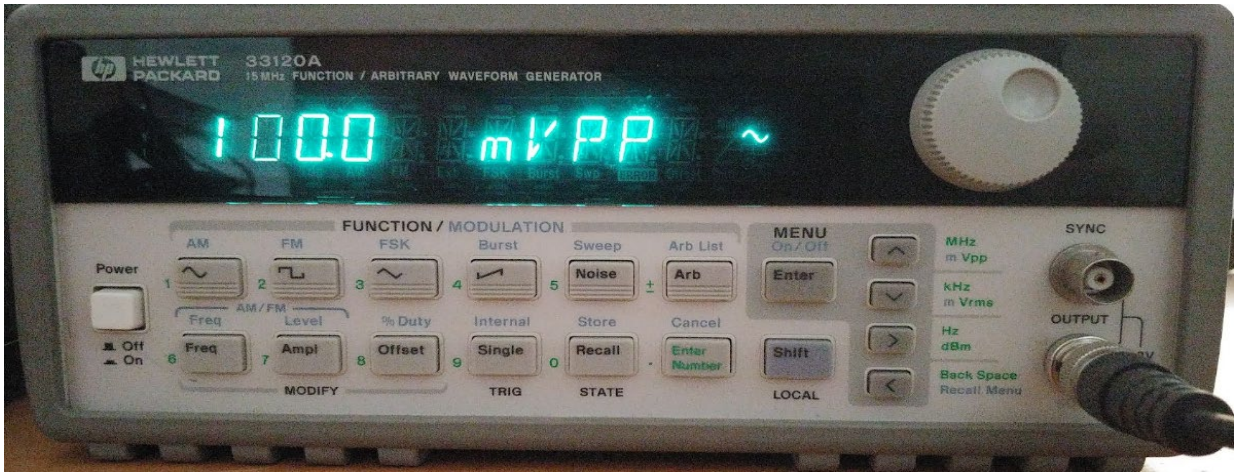
Figure 3.32 – Ammeter Display

4. Function Generator

Also known as a signal generator, the function generator provides different types of waveforms such as sine waves, square waves and triangle waves. The frequency and amplitude of theses waveforms can all be adjusted allowing for a wide range of signals to be created.

You tube introduction:

https://www.youtube.com/watch?v=PMHrgCdaEoc
https://www.youtube.com/watch?v=_pDz6e2ADew





Setting a waveform

Function generators typically have clearly marked waveform selections. Simply press the type of waveform you need and it will highlight to confirm your selection (Fig. 4.10).



Figure 4.10 – Waveform Selection

Setting the Frequency

1. Select the **Freq** function.

   Note: Some of these buttons are dual purpose. A white box will highlight the value you have selected.

2. Use the number pad to enter a value.

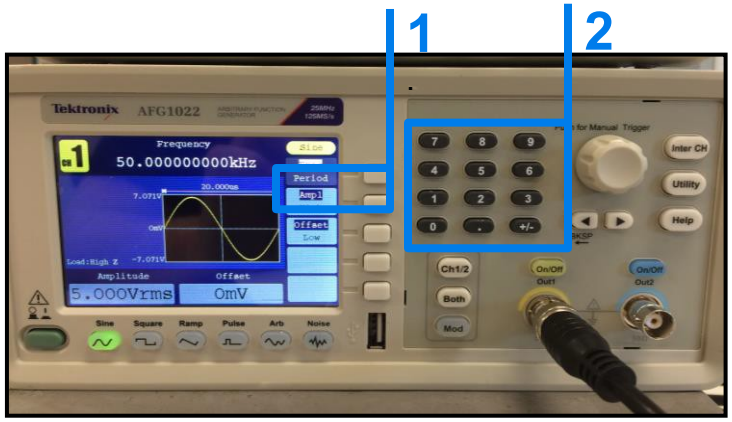


Figure 4.20 – Function Generator Display

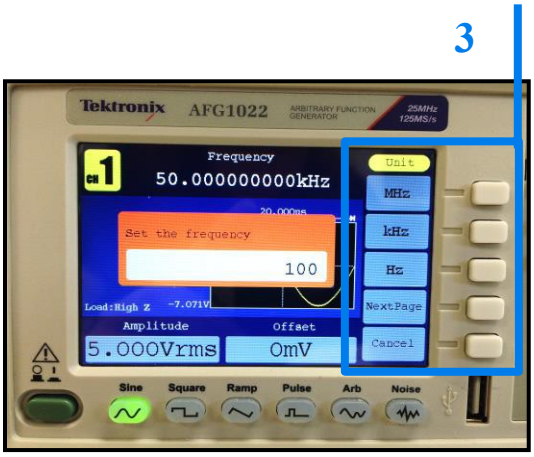


Figure 4.21 – Frequency Selection

3. Select the units for the value using the white buttons on right.

Note: Selecting the units will effectively hit 'enter' and the frequency will be set.

Setting the Amplitude

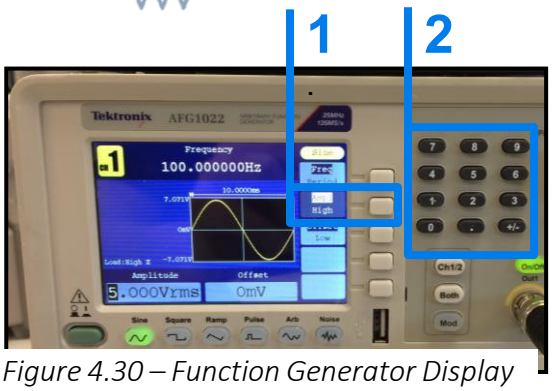


Figure 4.30 – Function Generator Display

1. Select the **Ampl** button.

2. Use the number pad first to select a numerical value.

3. Select the units for the numerical value fromthe options displayed at the right of the screen.

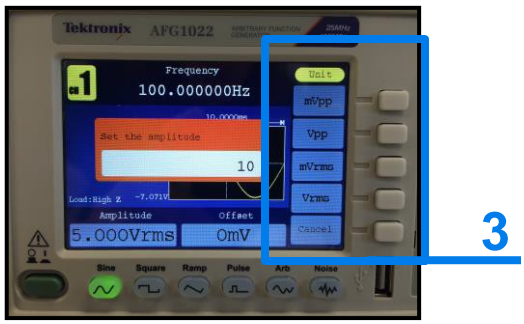


Figure 4.31 – Amplitude Selection

The Load and High Z

Often overlooked in the first few labs are the load function. The two main settings are High Z and 50 Ω. In almost every lab you will want to have the set as High Z.

The short explanation is that you are telling the generator that you are using a circuit with impedance other than 50Ω in series (the vast majority of cases). Using the 50Ω setting will cause the generator to produce double the amplitude that is set as output.



Figure 4.40 – Control Panel

1. Press the **Utility** button.

2. Select **Output Setup** on the display screen.

3. Press **CH1 Load.**

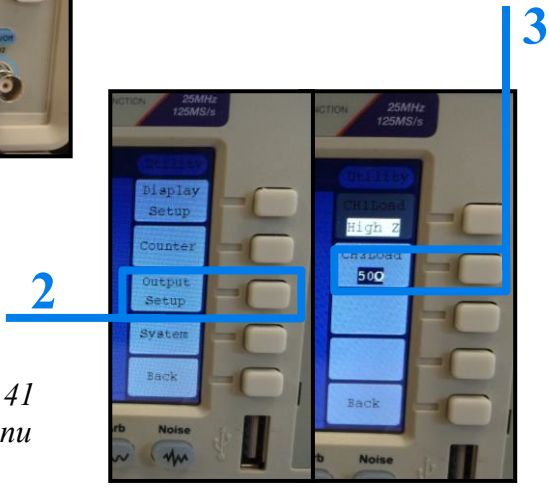


Figure 4.41 –Load Menu

Outputting a Signal

1. Connect the BNC connector into the channel jack on the function generator(Fig. 4.50).



Figure 4.50 – BNC Connection
The connectors will lock into the jack with a quarter turn.

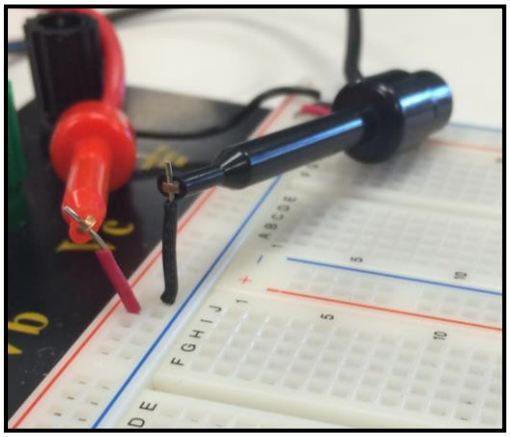


Figure 4.51 - Seizer Clips in the Power Rails

2. Place the red (+) and black (-) ends of theseizer cables across the circuit (Fig. 4.51).

Note: The seizer clips can be placed in the power rails as shown (fig 4.51) or by inserting them into the nodes corresponding to the power and ground of the circuit. If inserting them into the rails always check there is no other path for the current to travel other than your intended circuit!

3. Press the **ON/OFF**.

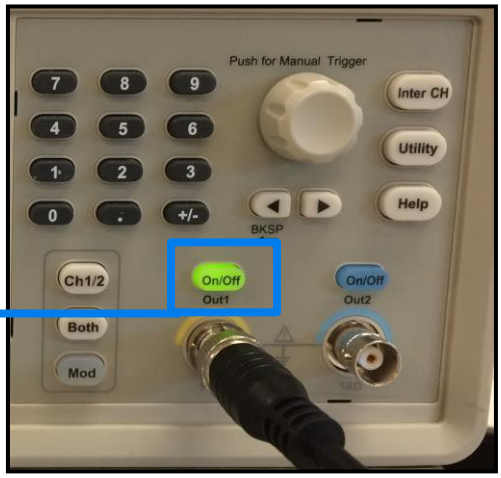


Figure 4.52 – Function Generator with the Output On

4. Oscilloscope

The oscilloscope is used to graphically depict voltage as a function of time. For most lab purposes these are generated by the function generator. All of the oscilloscopes in the labs have a minimum of two channels. This allows for multiple signals to be displayed on the screen. For our example we will only use a single channel.



Displaying a Waveform

1. Connect the BNC end of the oscilloscope probes to the oscilloscope jack (Fig. 5.10).

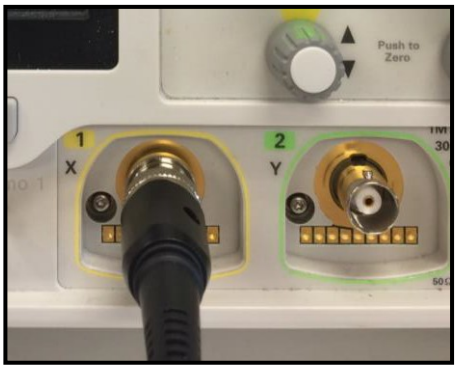


Figure 5.10 – Oscilloscope Connection

2. Connect the probe to your circuit (Fig. 5.11).

   Note: The probe will act as the point you are measuring from. The alligator clip will attach to ground reference. It can be attached to the same node as the function generator's seizure cables to measure over a whole circuit.

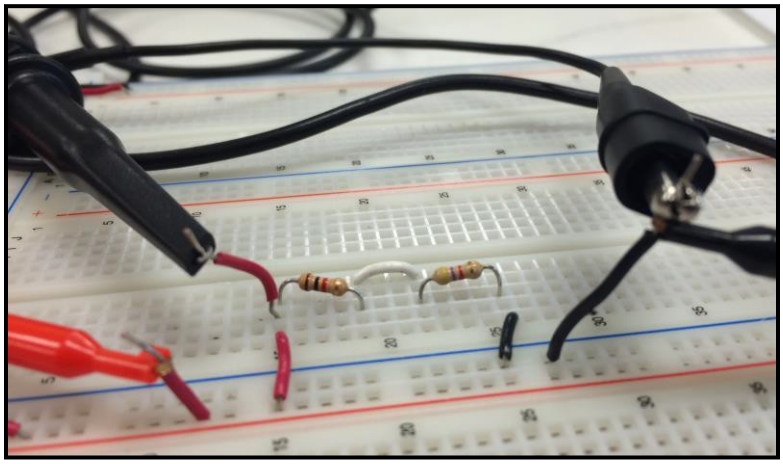


Figure 5.11 – Probe Setup for Circuit Measurement

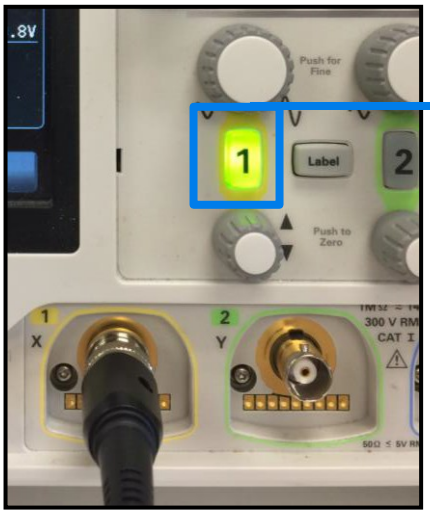


Figure 5.12 – Oscilloscope Output

3. Press **1** to output the signal.

Adjusting a Waveform

When it is first displayed the waveform will likely be unreadable. It may be clipped off-screen or bea thin line through the centre of the screen.
Knowing how to properly scale a waveform is acritical part of using the oscilloscope. As an example, figures 5.20 and 5.21 are the same waveform pre and post adjustment. We will provide a brief description of the basic adjustment controls to help you get started.

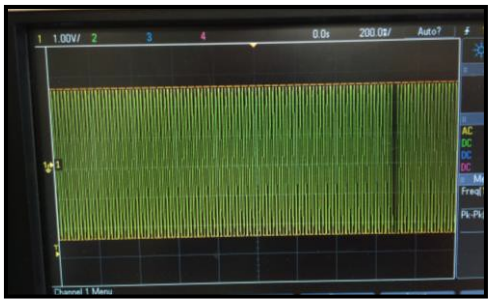


Figure 5.20 – Unadjusted Waveform

However, the best way to learn will be to spend some time adjusting different waveforms. Keep in mind that you are able to connect the function generator directly tothe oscilloscope with BNC to BNC cables to practice adjusting different waveforms without building a circuit.

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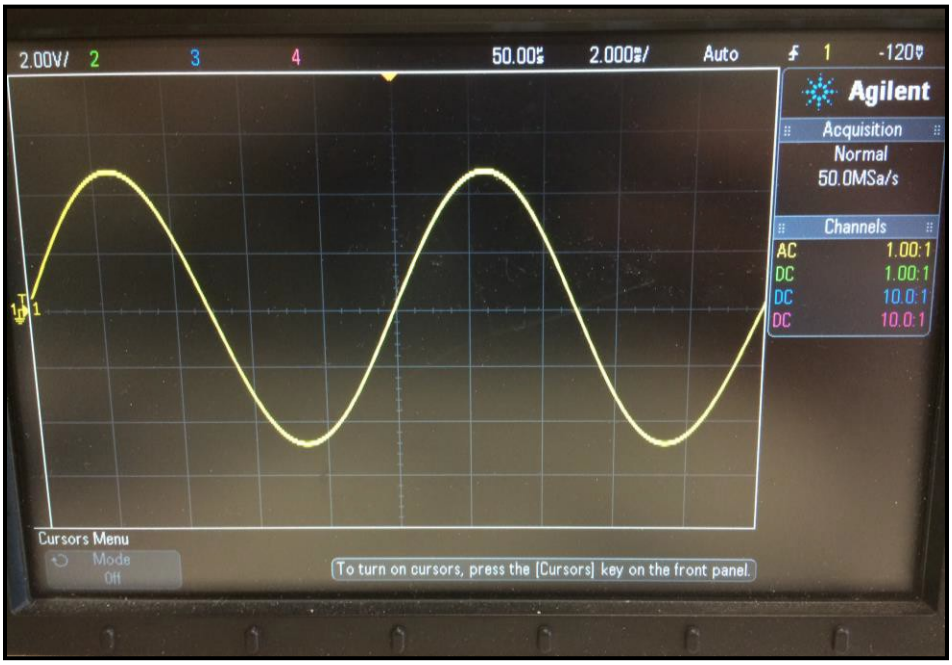


Figure 5.21 – Fully Adjusted Waveform
Keep in mind what your input signal is as you are making adjustments to help decide the range of values.

Adjustment Controls

The oscilloscope displays a graph with both horizontal and vertical divisions. Adjusting the vertical (amplitude) and horizontal (time) scales will adjust the value of each grid division. These values are generally displayed at the top of the screen. There is also an offset control that allows the signal to be moved up and down as a whole.

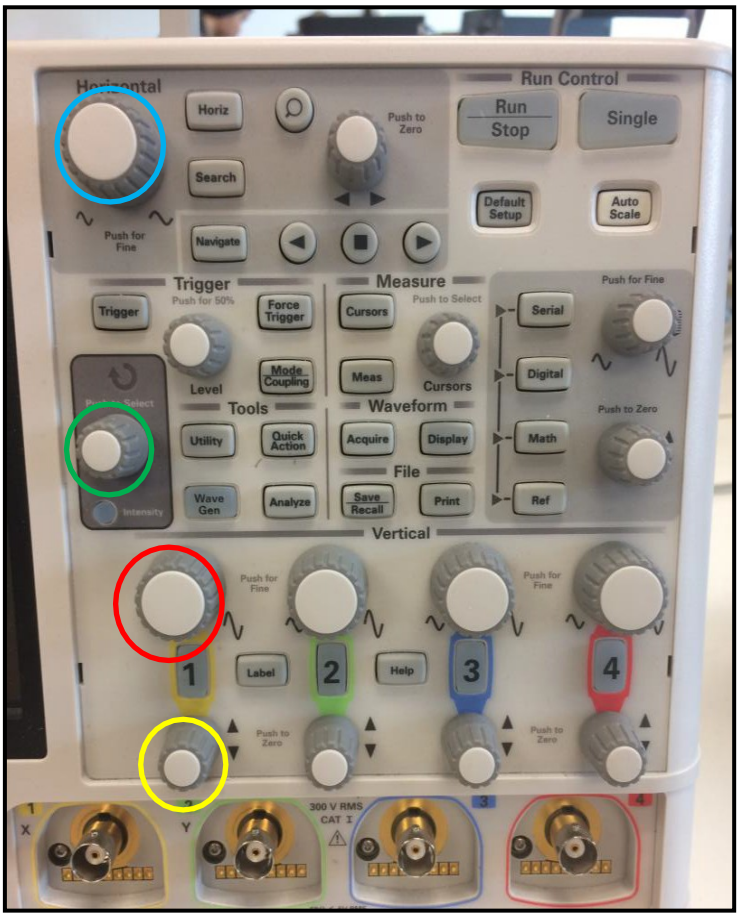
The three basic adjustment controls listed below are matched to figure 5.30. The time and amplitude controls can also be pushed in to switch to fine adjustments. The offset button can be pressed to 'zero' the waveform. The selection button has also been included as it is used to adjust many of the functions on the oscilloscope.

Time - Horizontal scale

Amplitude - Vertical scale

Offset - Shifts the waveform up or down

Selection - Menu control



Displaying Measurements

The oscilloscope is capable of displaying many measurements useful for circuit analysis; especially when comparing two wave forms. Most models are capable of displaying multiple measurements simultaneously. We will use peak to peak voltage as an example but all measurements can be found and displayed through the same steps.



Figure 5.40 – Measurement Controls

1. Press the **Meas** button located in the centre of the oscilloscope control panel.

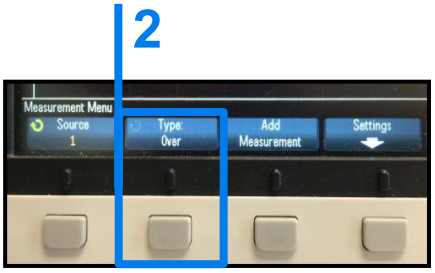


Figure 5.41 – Measurement Menu

2. Select **Type** using the push-button at the bottomof the oscilloscope display.

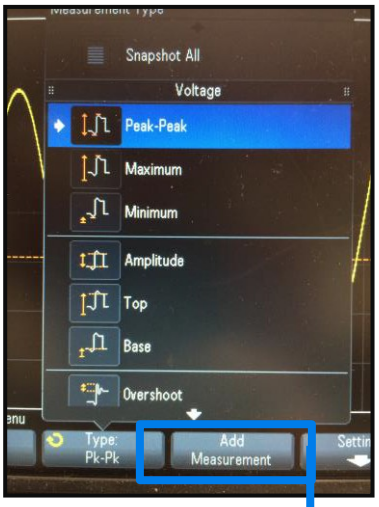


Figure 5.42 – Types of Measurements

3. Choose a measurement from the pop-up menu(Fig. 5.42).

4. Press the **Add Measurement** button.

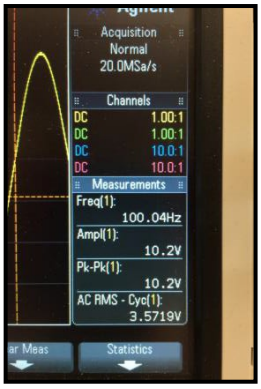


Figure 5.43 – Displayed Measurements

The measurement will now be displayed in the far right of the oscilloscope display (Fig. 5.43).It is possible to clear the measurements using the Clear Measurements function. Using the channel select you may also put up measurements for two different waveforms

Probe Settings

The oscilloscope probes have a red selector switch labelled 1X, 10X, and REF. REF is the ground reference position of the probe. The 1X and 10X refer to the probe's attenuation. In practical terms it refers to whether the signal will be 1:1 or magnified 10:1. The latter is usually used for higher accuracy and higher frequency measurements. For most purposes in the lab 1X will be suitable. However, the reason for the addition to this manual is that if unchecked this function can play tricks on your analysis.



Figure 5.50 – Probe Modes

As you may have noticed the oscilloscope display shows a 1:1 and 10:1 ratio. These directly related to the signal measured by the probe and must match the probes ratio and vice versa.

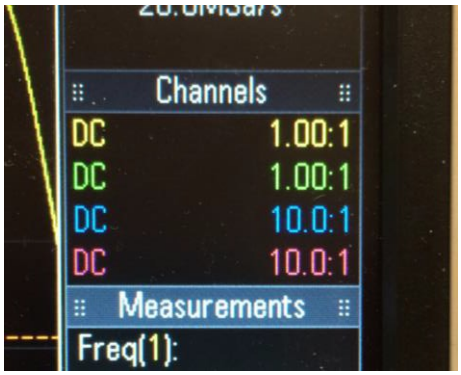


Figure 5.51 – Channel Ratios

Here is an example of the measurements for a ~10Vpp 100 hz sinewave with a 1:1 setting on the probe and oscilloscope.

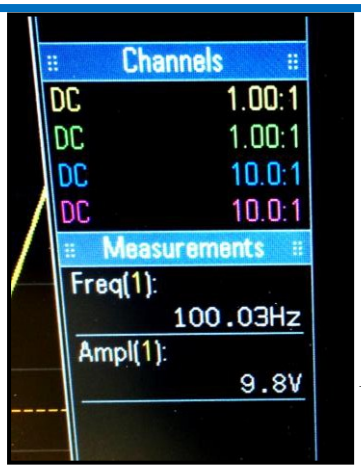


Figure 5.52 – 1 to 1 Ratio

Now the same input signal and probe setting but with the 10:1 ratio mistakenly left on.

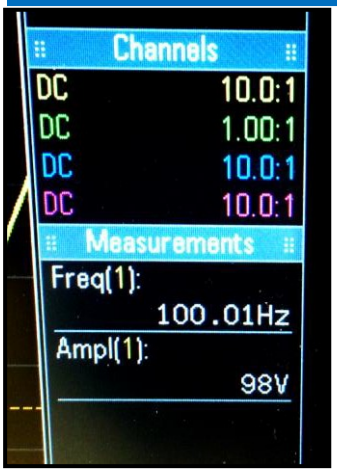


Figure 5.53 – 10 to 1 Ratio

Setting the Probe Ratio

1. Press the **1** button above the BNC connector.

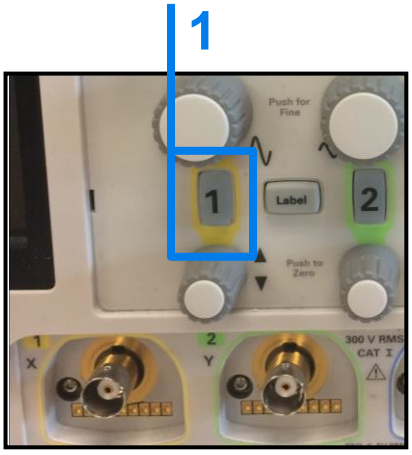


Figure 5.60 – Channel Selection

2. Select the **Probe** menu.



Figure 5.61 – Probe Menu

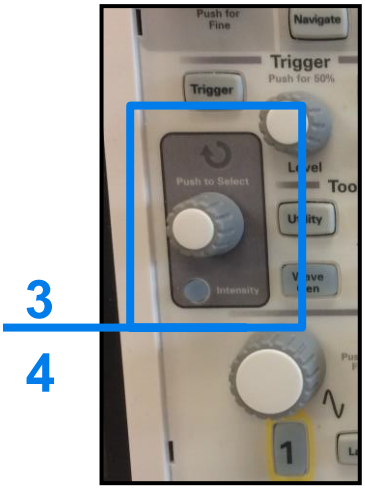
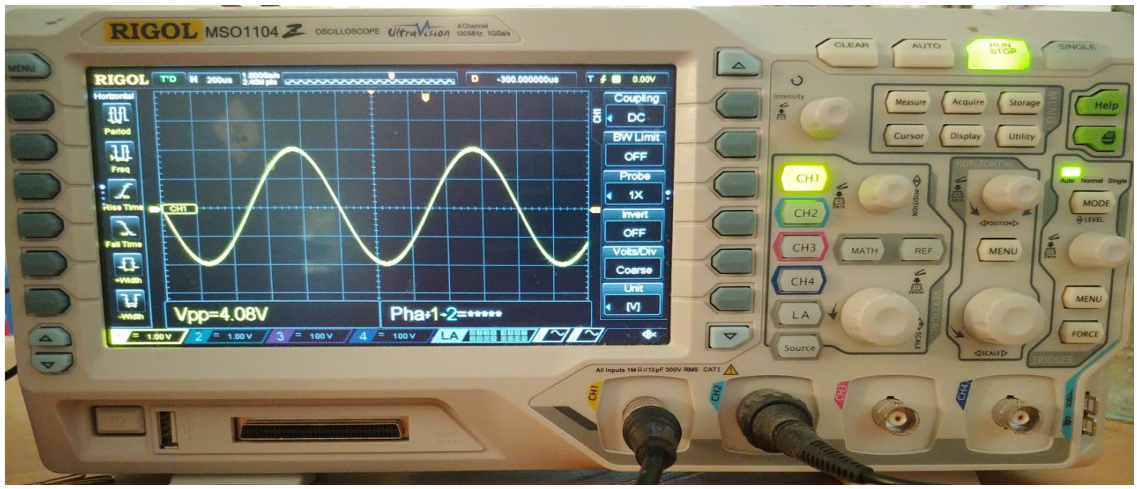


Figure 5.62 – Selector Control

3. Use the **Selector** knob to adjust the attenuation.

4. Push the **Selector** knob in to apply the setting.

RIGOL Oscilloscope:



Some links:
https://learn.sparkfun.com/tutorials/how-to-use-an-oscilloscope/all Sparkfun tutorial using a scope – available as a pdf.

https://www.rigolna.com/basics-of-oscilloscopes/ Introduction to oscilloscopes pdf from Rigol

https://www.youtube.com/watch?v=Ogyk_zwWujE 10-minute overview of the Rigol scope on YouTube

We can have Analog Oscilloscopes (much rarer now), Digital Storage Oscilloscopes (DSO), and Mixed Signal Oscilloscopes (MSO) which will have 2 – 4 analog inputs and many digital inputs.

The Display Grid

In the photo below, using the scaling indicated on the screen, you should be able to determine the amplitude and frequency of the signal shown.



NOTE: for more details on the following functions and features of the Rigol scopes in 3 (4?) of our labs (TECH 227, TEC 204, TEC 205, and CBA 119?) see Tom Briggs, *Introduction to Oscilloscopes - A Hands-On Laboratory Guide to Oscilloscopes using the Rigol DS1104Z*
The pdf is here: https://www.rigolna.com/basics-of-oscilloscopes/.

Key Points for Oscilloscope and Function Generator

1. Auto scale is a wonderful thing, but take the time to become confident with the V/div and sec/div controls, as auto scale will not work for low frequency and extremely noisy signals.

2. Voltage amplitude measurements can be done automatically (peak, peak-to-peak, rms, average), as can time measurements (period, frequency, delay, phase delay). Observe the screen carefully to see what cursor positions the scope is using for its calculations.

   Sometimes the peak-to-peak values of noisy signals will be far too large because minimum and maximum values are used. Phase delays may be reported inconsistently.

3. When in doubt, use voltage and time cursors manually to get good data.

4. When your scope is reporting voltages that are 10 x or 1/10 x the amplitude you set on your function generator, check that the slider on the probe matches the scope probe setting for that channel.

5. When your scope is reporting voltages that are half the amplitude you set on your function generator (especially on the HP33120A, used in CBA 119, CBA 124 and TEC 229), then the function generator's output termination is set incorrectly. In the SYS-OUT TERM menu, there are two choices: 50 ohms or HIGH-Z. Since the internal resistance is 50 ohms, the 50 ohms output termination results in a halving of output voltage. Choose HIGH-Z to ensure the voltage amplitude you dial in is also the voltage amplitude applied to your circuit and measured on the scope.

6. If your scope display moves, you may be able to adjust triggering source, triggering method, or triggering level to stop it. If this fails, click on Run-Stop to immobilize your display.

7. The ground clips on your oscilloscope probes are earth grounds. Therefore, when you are using both channels of the scope, your ground clips must be electrically in the same place. If they are not, you are shorting out a part of your circuit.

8. Normally you should use DC coupling to view your signals. If you wish to remove the DC offset from a signal (to examine the AC portion more carefully), change to AC coupling.

9. To capture an image from your scope for insertion into a Word document:
   a. Make sure your scope is physically connected to COM1 of your computer.
   b. On the scope, push the Utility button to get the I/O menu. Make sure the baud rate is 19200 baud.
   c. Double-click on the Scope Capture Word 2003 icon on your desktop.
   d. In Word, click on the Add-Ins menu. From the scope capture toolbar, click on the Connect to Scope icon.
   e. Click on Search. The program should find your scope on COM1. Click OK twice.
   f. From the scope capture toolbar, click on the camera icon to insert your scope image into your document. Click OK. The transfer process will take between 10 and 90 seconds.

Final Notes

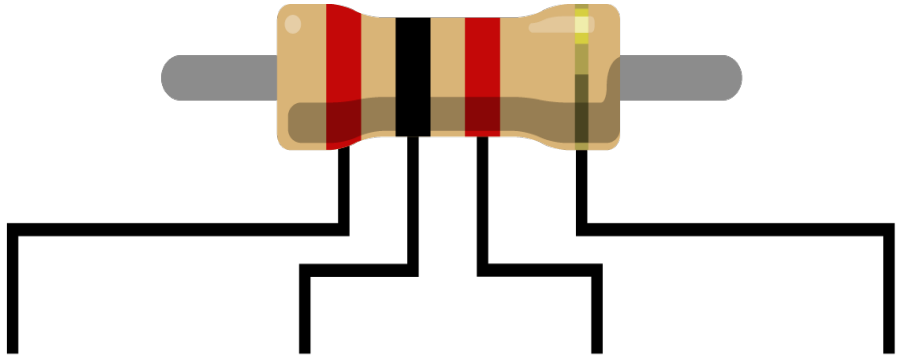
We hope that you now have a basic understanding of the instruments used in circuit analysis. In the appendices we have added an *additional resources* section that includes some websites we have found helpful for electronics courses.

The make and model for each device used in our examples have also been included to help find their data sheets online.

Happy circuiting!

Appendix

Appendix A -Resistor Colour Code Chart



1st digit	2nd digit	Multiplier	Tolerance
0	0	x 1	
1	1	x 10	±1%
2	2	x 100	±2%
3	3	x 1K	
4	4	X 10K	
5	5	x 100K	
6	6	x 1M	
7	7		
8	8	x 0.1	±5%
9	9	x 0.01	±10%

Appendix B – Additional Resources

Helpful Websites:

- All About Circuits – www.**allaboutcircuits**.com
- Hyper Physics – hyperphysics.phy-astr.gsu.edu/hbase/emcon.html#emcon

Devices Makes and Models:

Devices:
Digital Multimeter: HP 34401A
Oscilloscope: Agilent Tech Infiniivision MSO-x3014A
Power Supply: GW Instek GPD-3303S
Function Generator: Tektronix AFG1022

Power Supply: XANTREX
Function Generator: Hewlett Packard 33120A
Oscilloscope:: Rigol mso1104